Capital University of Science and Technology, Islamabad
Department of Electrical and Computer Engineering

EEG1611: Applications of Information and Communication Technologies Lab

Pre-requisite(s):	None
Credit Hours:	1
Total No. of Labs:	12+2
Recommended Book(s):	Lab Manual

Course Description:

This lab covers the basic knowledge of commonly used information and communication technologies (ICT) tools including word, excel, power point and Visio. It also provides an overview of computer networks along with various network simulator tools. Furthermore, this helps the students to develop algorithms, flowcharts and provides basic knowledge of communication protocols and database management systems.

Course Learning Outcomes:

At the end of this course, the students should be able to:

No.	Description	Taxonomy Level	PLOs
CLO:1	Explain the basic concepts, features and working principles of ICT tools, algorithms, programming constructs, computer networks, databases and information security.	C2	PLO:1
CLO:2	Arrange the software environment, tools and settings required to carry out assigned ICT, programming and networking tasks.	P2	PLO:5
CLO:3	Practice the use of ICT tools, programming constructs, network configurations and database operations by following prescribed procedures to produce documents, spreadsheets, presentations, flowcharts, programs and database objects.	P3	PLO:5
CLO:4	Communicate the outcomes of ICT-based tasks and the small-scale project, using structured documentation and spreadsheet tools, during a viva assessment.	A2	PLO:5

List of Experiments:

1- Exploring Computer Hardware and Software in Practice

2- Creating and Editing Professional Documents in Microsoft Word

3- Building Spreadsheets, Formulas, and Charts in Microsoft Excel

4- Analyzing Data with Advanced Excel Functions

5- Designing Effective Presentations in Microsoft PowerPoint

6- Drawing Block Diagrams and Flowcharts in Microsoft Visio

7- Writing Pseudocode and Designing Flowcharts for Problem Solving

8- Setting Up Visual C++ and Running Your First Program

9- Making Decisions in Code with Relational and Logical Operators

10- Familiarization with Network Simulation Tools

11- Designing and Querying a Database in Microsoft Access

12- Protecting Systems with Antivirus, Firewalls, and Encryption

13- Open-ended Lab I

14- Open-ended Lab II